

Bangyan Liao

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Education

Hunan University, BS in Automation.	Sept 2019 – June 2023
Westlake University & Zhejiang University (Joint Program), PhD in Computer Science and Technology. Advisor: Prof. Peidong Liu.	Sept 2023 – June 2028 (expected)

Research Interests

Multi-view Geometry Problems in 3D Computer Vision	2023 - 2024
Optimization :	2023 -
Globally Optimal Outlier-Robust Optimization	
Learning to Globally Optimal Optimize	
Learning to Transport, Inference and Control Probability Path Measures :	2024 -
[Transport] Optimal Transport / Schrödinger Bridge	
[Control] Stochastic Optimal Control / Mean-Field Control	
[Inference] Nonlinear Smoothing / Filtering	

Publications

Advances in Global Solvers for 3D Vision	2026
Z Zhao, H Yang, B Liao , Y Zeng, S Yan, Y Gu, P Liu, Y Zhou, H Li, J Civera	
Under Review	
scDFM: Distributional Flow Matching Model for Robust Single-Cell Perturbation Prediction	2026
C Yu*, C Wang*, B Liao , T Wu	
International Conference on Learning Representations (ICLR)	
Neural Predictor-Corrector: Solving Homotopy Problems with Reinforcement Learning	2026
J Mai*, B Liao *, Z Zhao, Y Zeng, H Li, J Civera, T Wu, Y Zhou, P Liu,	
International Conference on Learning Representations (ICLR)	
E-MoFlow: Learning Egomotion and Optical Flow from Event Data via Implicit Regularization	2025
W Li*, B Liao *, Z Yi, Q Xu, P Wu, P Liu,	
Advances in Neural Information Processing Systems (NeurIPS)	
DaCapo: Score Distillation as Stacked Bridge for Fast and High-quality 3D Editing	2025
Y Huang, B Liao , Y Hu, H Lin, L Wu, S Li, C Tan, Z Liu, Y Liu, Z Zang, C Yu, Z Lei	
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)	
Convex Relaxation for Robust Vanishing Point Estimation in Manhattan World	2025
B Liao *, Z Zhao*, H Li, Y Zhou, Y Zeng, H Li, P Liu	
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)	
[Full Score] [Oral][Award Candidate]	
EvTTC: An Event Camera Dataset for Time-to-Collision Estimation	2025
K Sun*, J Li*, K Dai, B Liao , W Xiong, Y Zhou	
IEEE Robotics and Automation Letters (IEEE RAL)	
Ego-motion Estimation for Vehicles with a Rolling Shutter Camera	2024
Y Zhang, B Liao , D Qu, J Wu, X Lu, W Li, Y Xue, Y Lao	
IEEE Transactions on Intelligent Vehicles (IEEE TIV)	
RSL-BA: Rolling Shutter Line Bundle Adjustment	2024
Y Zhang*, B Liao *, Y Xue, C Lu, P Liu, Y Lao	

European Conference on Computer Vision (ECCV)	
GlobalPointer: Large Scale Plane Adjustment with Bi-Convex Relaxation	2024
<i>B Liao*</i> , Z Zhao*, L Chen, H Li, D Cremers, P Liu	
European Conference on Computer Vision (ECCV)	
Motion and Structure from Event-based Normal Flow	2024
Z Ren*, <i>B Liao*</i> , D Kong, J Li, P Liu, L Kneip, G Gallego, Y Zhou	
European Conference on Computer Vision (ECCV)	
Event-Aided Time-to-Collision Estimation for Autonomous Driving	2024
J Li*, <i>B Liao*</i> , X Lu, P Liu, S Shen, Y Zhou	
European Conference on Computer Vision (ECCV)	
USB-NeRF: Unrolling Shutter Bundle Adjusted Neural Radiance Fields	2024
M Li*, P Wang*, L Zhao, <i>B Liao*</i> , P Liu	
International Conference on Learning Representations (ICLR)	
Fast Rolling Shutter Correction in the Wild	2023
D Qu*, <i>B Liao*</i> , H Zhang, O Ait-Aider, Y Lao	
IEEE Transactions on Pattern Analysis and Machine Intelligence (IEEE TPAMI)	
Revisiting Rolling Shutter Bundle Adjustment: Toward Accurate and Fast Solution	2023
<i>B Liao*</i> , D Qu*, Y Xue, H Zhang, Y Lao	
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)	

Awards

CVPR 2025 Award Candidate (15/13008 submissions)	Aug 2025
National Scholarship	Oct 2025
National Natural Science Foundation of China (NSFC) PhD Program	Dec 2025
China3DV 2026 Rising Star Award	Apr 2026

Invited Talks

Convex Relaxation for Robust Vanishing Point Estimation in Manhattan World	2025
Oral Presentation	
CVPR Conference, Nashville, TN, USA.	
Global Solvers for 3D Vision: From Classical Convex Relaxation to General Homotopy Paradigms	2026
Oral Presentation at the Outstanding Student Forum	
China3DV Conference, Hangzhou, China.	

Experience

Visiting Student , AI4S Lab, Westlake University	Feb 2025 –
- Advisor: Prof. Tailin Wu. - Researched at fundamental machine learning algorithm.	
Visiting Student , Prof. Lao's Lab, Hunan University	Apr 2021 – June 2023
- Advisor: Prof. Yizhen Lao. - Researched at novel fundamental geometric and optimization solvers on Rolling Shutter camera. - Resulted in IEEE TPAMI and CVPR 2023 paper.	
Visiting Student , NAIL, Hunan University	Nov 2021 – June 2023
- Advisor: Prof. Yi Zhou. - Researched at novel fundamental geometric and optimization solvers on Event camera. - Built and developed a vehicle forward warning software and hardware system with Event camera.	

- Resulted in two ECCV 2024 papers.

Student Mentoring

Yingping Zeng	Aug 2024 -
• co-supervised master student with Prof. Yi Zhou from Hunan University	
Hao Li	Oct 2024 - Apr 2025
• visiting undergraduate student from Shandong University	
Zhisheng Song	Sep 2024 -
• visiting undergraduate student from Shandong University	

Reviewing

The IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR)	2023 -
The International Conference on Computer Vision Conference (ICCV)	2024 -
The AAAI Conference on Artificial Intelligence (AAAI)	2024 -
The ACM Multimedia (ACM MM)	2025 -
The International Conference on 3D Vision (3DV)	2024 -
The IEEE International Conference on Robotics and Automation (ICRA)	2024 -
The IEEE Robotics and Automation Letters (RAL)	2024 -
The Conference on Neural Information Processing Systems (NeurIPS)	2025 -
International Conference on Learning Representations (ICLR)	2025 -

Endurance Sports

Trail Donghai (29 km, 1207m D+, ITRA 1)	2026
Finish Time: 05:42:37	
Xihu Qingzhiwu Longdong Trail (32 km, 1723m D+, ITRA 2)	2025
Finish Time: 07:12:13	
The North Face 100 Challenge Moganshan (33 km, 1530m D+, ITRA 2)	2019
Finish Time: 04:05:08	
Mount Siguniang Meconopsis Trail (29 km, 1170m D+, ITRA 1)	2019
Finish Time: 06:16:17	
Westlake Half Marathon	2026
Finish Time: 01:56:13	
HLA Round Lihu Lake Half Marathon	2025
Finish Time: 01:49:58	
Westlake University 3rd Campus Half Marathon	2025
Finish Time: 02:03:04	
Wenjiang Half Marathon	2018
Finish Time: 01:42:29	